Internet of Things

Module 1: INTRODUCTION & DOMAIN APPLICATIONS

1. Introduction to Internet of Things (IoT)

The IoT refers to the interconnection of everyday physical objects through the internet, enabling them to collect and exchange data. These objects ranging from household appliances to industrial machines are embedded with sensors, software, and other technologies that allow them to send and receive data, enhancing their functionality and automation. The **IoT** is also defined as a transformative concept that facilitates communication among internetworked devices and applications. In this paradigm, physical objects referred to as "things" are embedded with sensors and communication capabilities, enabling them to interact and exchange data over the Internet.

IoT originated from the need to connect physical objects with digital systems to improve automation, monitoring, and data-driven decision-making. One early example is the **Radio Frequency Identification Device (RFID)**, which is used as an identity communication tool. RFID tags allow objects to be uniquely identified and tracked remotely via computers connected through the Internet. IoT enables smarter decision-making, real-time monitoring, and efficient processes across various domains, including healthcare, transportation, agriculture, and home automation.

IoT encompasses a wide range of applications, such as: **GPS-based tracking systems** for location monitoring, **Machine-to-Machine (M2M) communication** for industrial automation, **connected cars** for real-time vehicular data exchange, **Wearable technologies** that communicate with smartphones, and **Industry 4.0** implementations enabling smart factories. The IoT revolution has contributed significantly to the development of smart cities and is anticipated to play a vital role in the evolution of self-driving vehicles. The subsequent sections provide further insights into IoT fundamentals, including its definition, vision, conceptual framework, and architectural design.

Definition:

Internet is a global network of computers, mobiles, and tablets connected via servers, governed by standard protocols. It enables sending, receiving, and communication among connected devices. Whereas “thing” can be any object, such as a bicycle, or an umbrella. Therefore the ‘Internet of Things’ can be defined as a network of physical things sending, receiving, or communicating information using the internet or other data networks.

IoT Conceptual Framework:

An umbrella can be transformed into an IoT device, it can be connected to internet specifically to the weather forecasting websites and regular updates can be shared with the user via different colored LEDs or mobile notifications. The IoT framework defines the systematic integration of physical devices with network connectivity and data analytics. It can be understood using the following equation:

Physical object controller, sensor and actuators + Internet = IoT

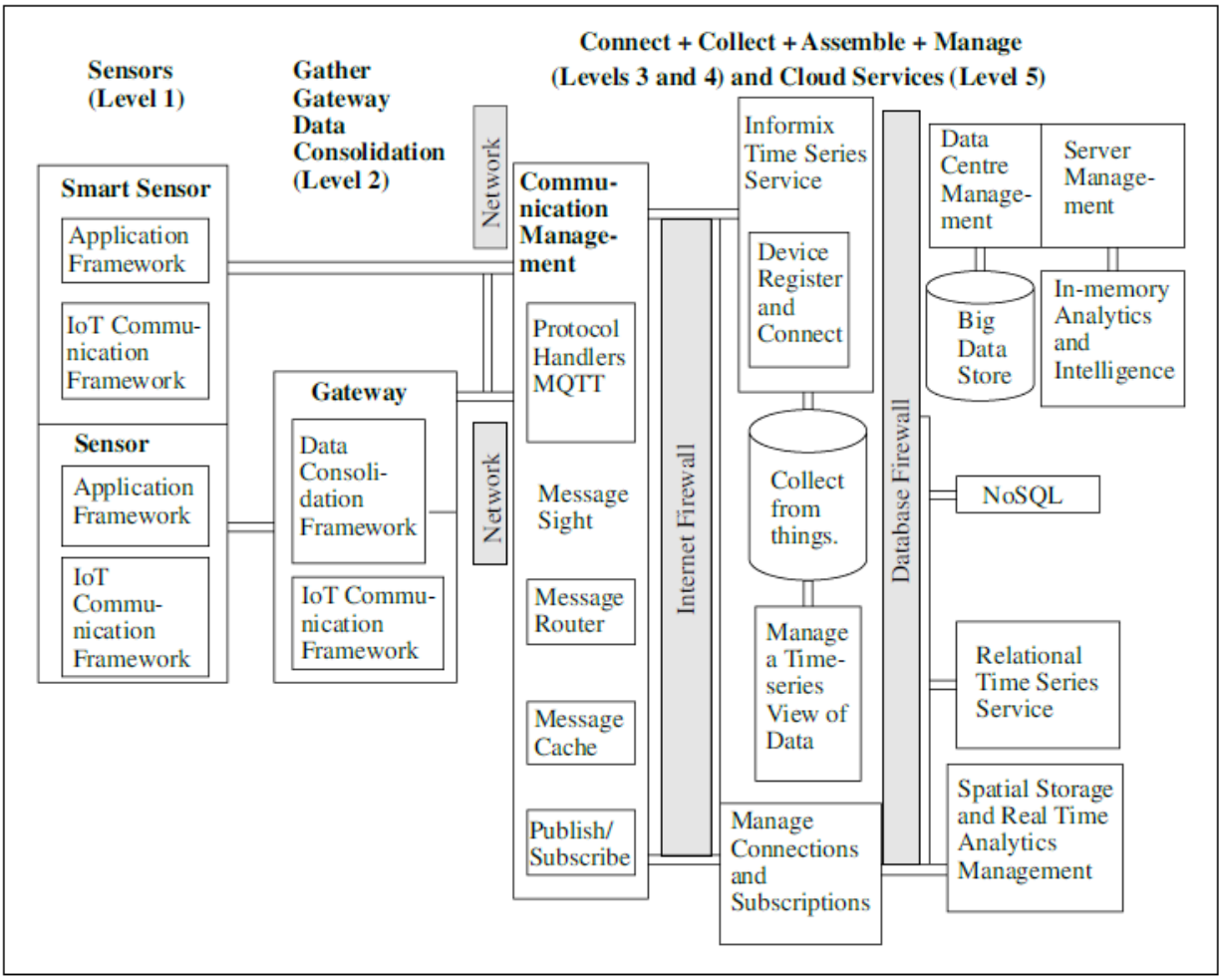


Figure 1.1: IoT conceptual framework.

In IoT frameworks there are multiple layers such as: **Sensing Layer**: Includes all sensors and devices that detect physical parameters such as temperature, humidity, motion, etc. Decoding the data before data transfer between two entities in another layer. Then comes the next layer, **Network Layer**: Responsible for the communication of sensed data to central systems through technologies like Wi-Fi, 5G, or LPWAN. **Data Processing Layer**: Involves storage, filtering, and analysis of data, often using cloud or edge computing. **Application Layer**: Where the data is transformed into meaningful information and made accessible to end-users through user interfaces. To represent the actions and communication of data at successive levels in IoT, the following equations can be defined.

Gather+ Enrich+ Stream + manage + Acquire + Organize and Analyze = IoT with connectivity to data center, enterprise of cloud server

If the IoT services usees data from the internetwork or cloud services, then the framework will be different,

Gather+ Consolidate + Connect + Collect + Assemble + Manage and Analyze = IoT with connectivity to cloud services

IOT Architectural View:

The general architecture of IoT consists of: - **Perception Layer**: The physical layer that senses and gathers data from the environment. This includes sensors, RFID tags, and actuators. **Network Layer**: Transfers the collected data using communication protocols like MQTT, CoAP, or HTTP over TCP/IP. **Middleware Layer**: Processes the data and manages devices. It includes storage systems, analytics platforms, and cloud services. - **Application Layer**: Delivers specific services like smart home control, health monitoring, etc. This layered architecture ensures scalability, flexibility, and interoperability in IoT solutions.

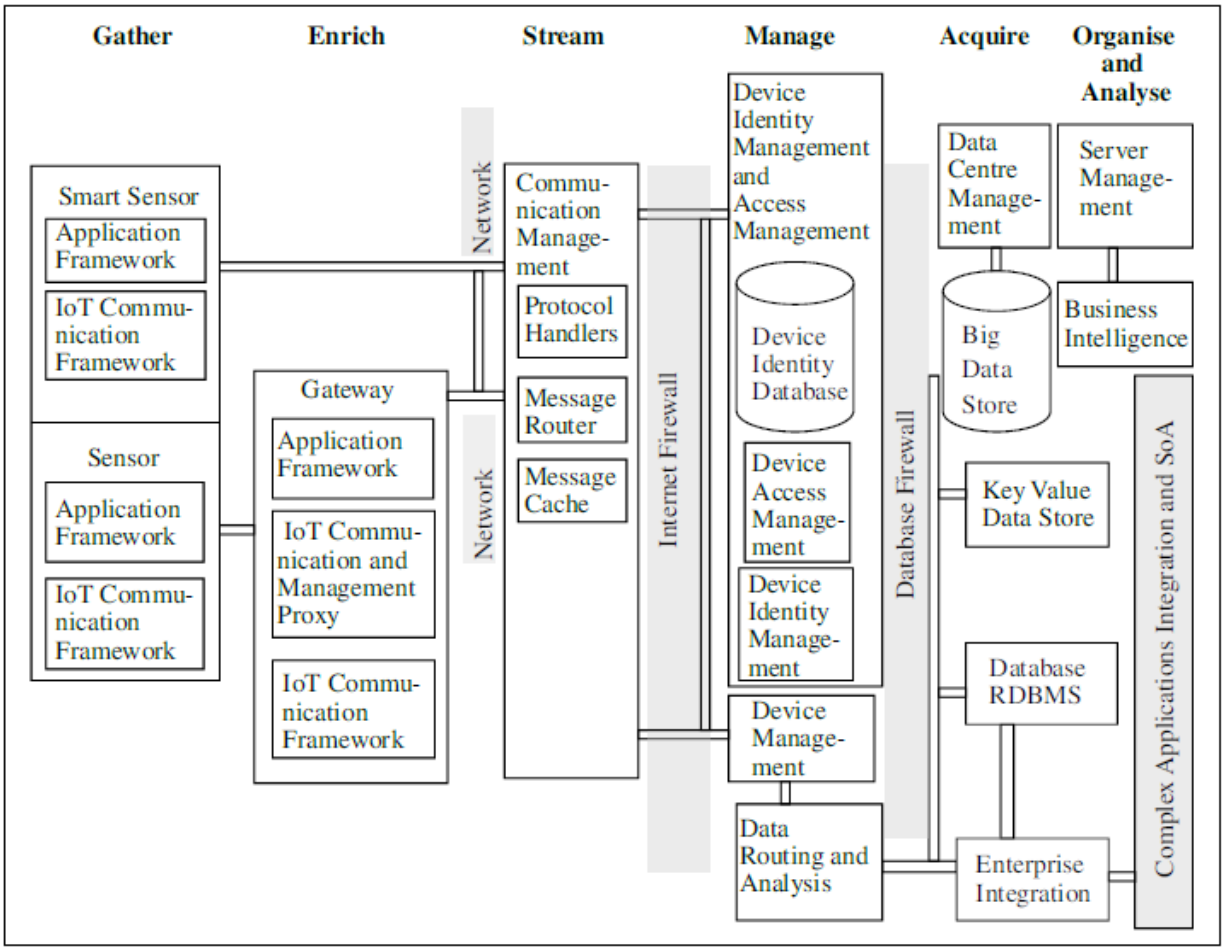


Figure 1.2: IoT architecture by Oracle.

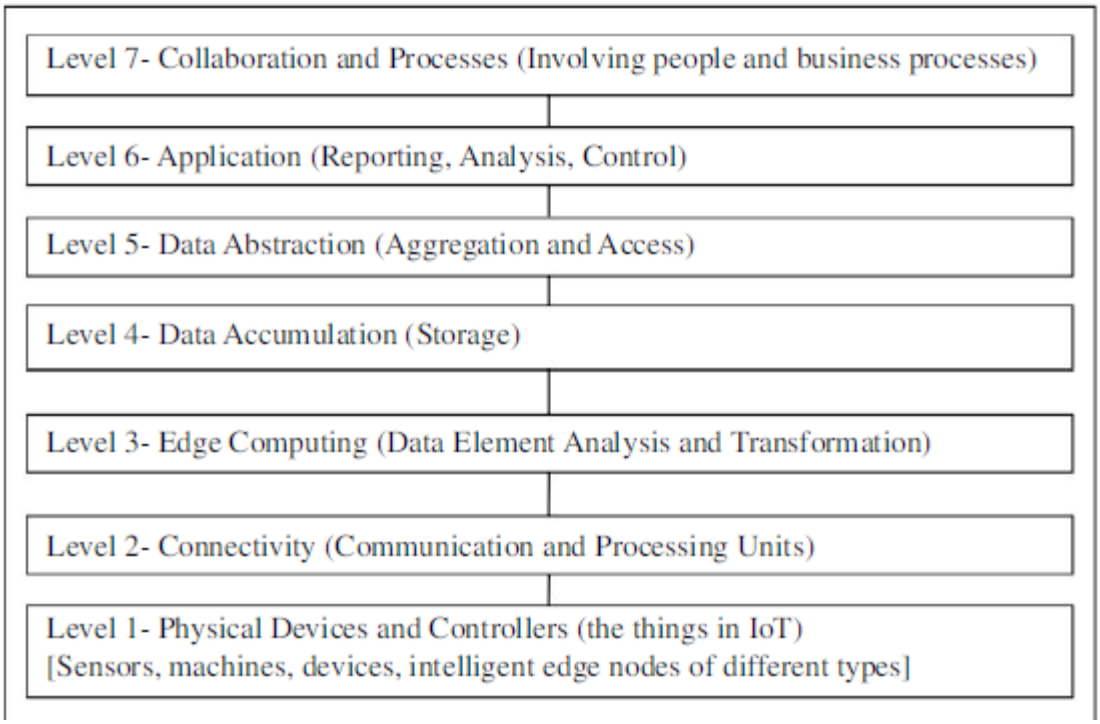


Figure 1.3: IoT reference model by CISCO.

2. Technology Behind IoT

IoT is a convergence of various technologies, including:

- **Embedded Systems and Microcontrollers**: Devices like Arduino, Raspberry Pi, and ESP32 that serve as the brains of IoT endpoints.
- **Sensors and Actuators**: Capture data and perform actions based on received commands (e.g., turning lights on or off).
- **Cloud Computing**: Offers scalable data storage, processing, and analytics. Examples include AWS IoT, CISCO IoT, BlueMix, openHAB, EvryThng, and Google Cloud IoT.
- **Big Data Analytics**: Extracts actionable insights from vast amounts of generated data.
- **Artificial Intelligence and Machine Learning**: Enhance IoT systems with decision-making capabilities, predictions, and pattern recognition.
- **Cybersecurity**: Protects IoT ecosystems against data breaches, unauthorized access, and attacks through encryption and authentication mechanisms.
- **Integrated development environment (IDE)** for developing device software, firmware and APIs.
- **Communication:** Ethernet, RFID, NFC, UWB, ZigBee, Bluetooth, WiFi, WiMax, 2G/3G/4G.
- **Network backbone:** IPv4, IPv6, UDP, and 6LowPAN.

2.1 Server-end technology

IoT servers are enterprise servers, cloud servers, data centers and databases. These servers offer following components,

- Online platforms
- Devise identification, and management
- Data accruing, integration, organizing and analysing
- Use of web applications, services and business processes

2.2 Major components of IoT System

- Physical object with embedded software into a hardware.
- Hardware with a microcontroller, sensors and control unit.
- Communication module.
- Software.

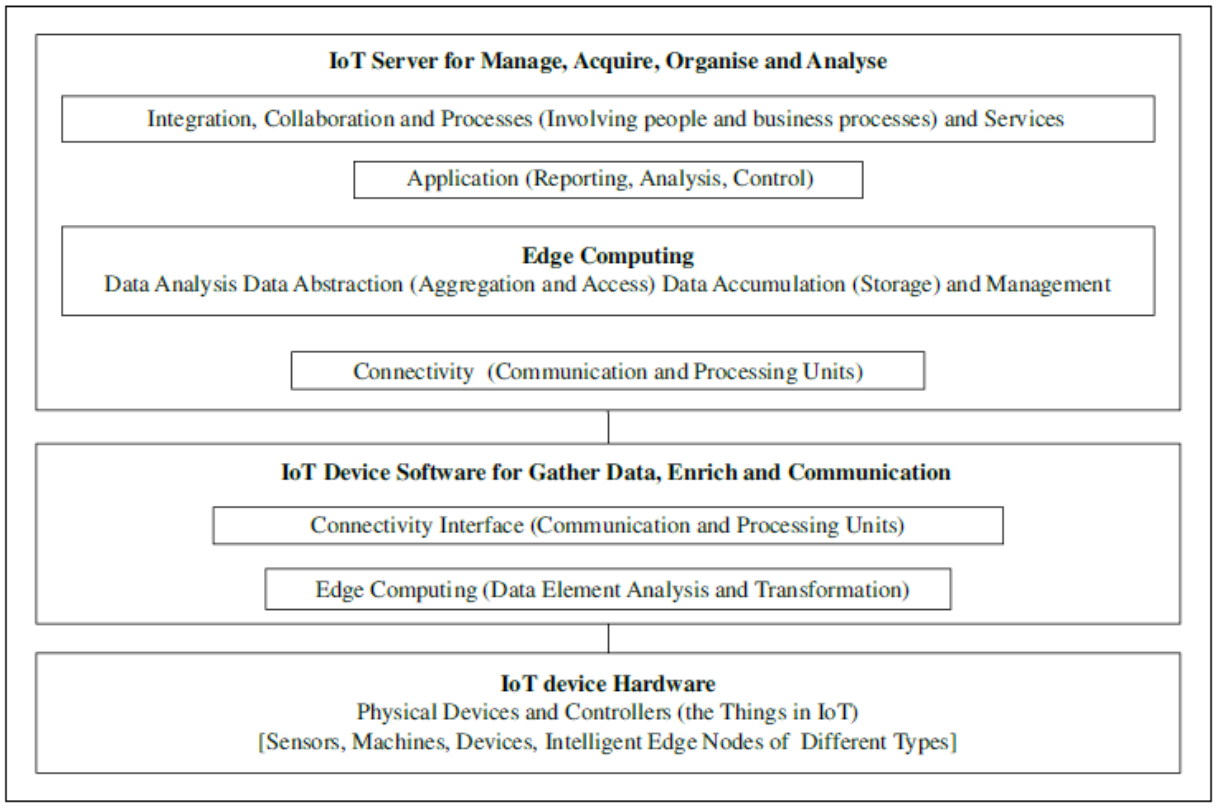


Figure 1.4: IoT components for device.

2.3 Development tools and open-source frameworks for IoT implementation

- Eclipse IoT
- Arduino
- Kinoma software platform

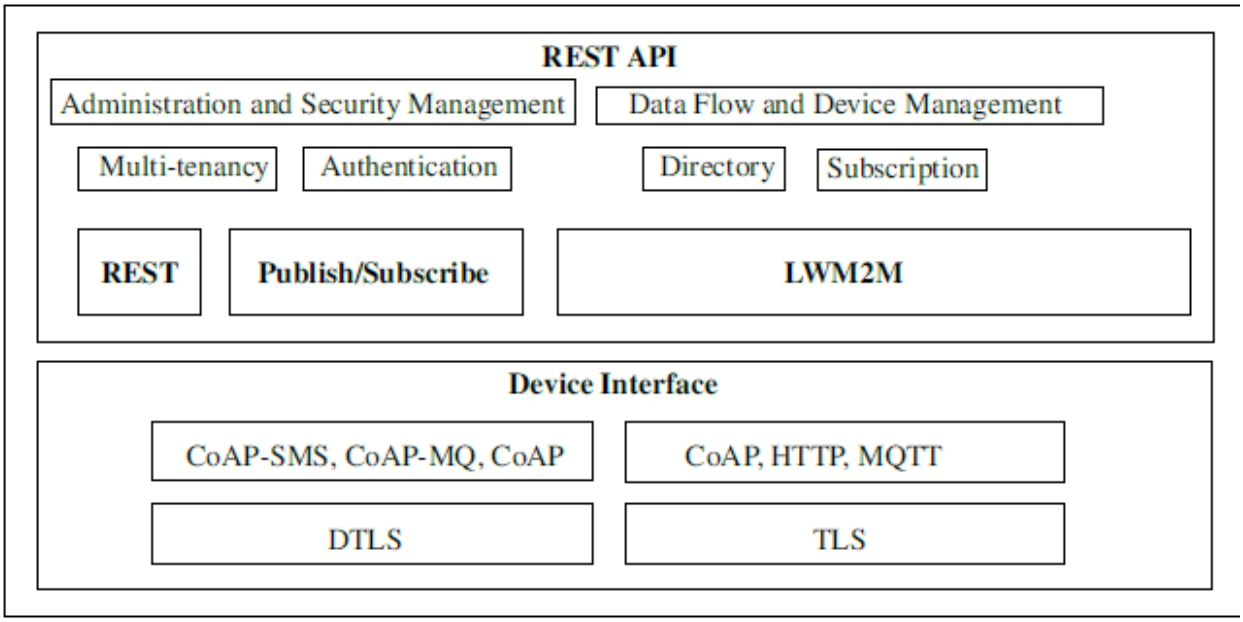


Figure 1.5: API and device interfacing components

3. Communication Technologies

Communication is vital in IoT systems, and the choice of technology depends on the application:

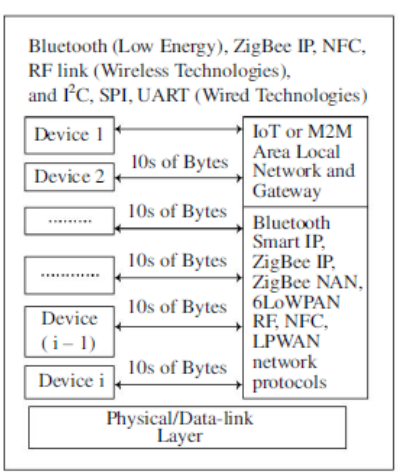
3.1 Wireless Communication Technology:

- **Short Range**:
  - **Bluetooth**: Ideal for personal devices (e.g., fitness bands).
  - **Zigbee**: Suitable for home automation with low power consumption.
  - **RFID**: Used for inventory tracking and access control.
  - **NFC (Near Field Communication)**: Used in payment systems and identification.
- **Medium Range**:
  - **Wi-Fi**: High data rate, suitable for home and office use.
  - **Z-Wave**: Used in smart homes for lighting and security.
- **Long Range**:
  - **LTE/4G/5G**: Provides wide-area coverage with high speed and bandwidth.
  - **NB-IoT (Narrowband IoT)**: Optimized for low power and long battery life.
  - **LoRaWAN (Long Range Wide Area Network)**: Ideal for applications requiring wide-area coverage and low data rates.

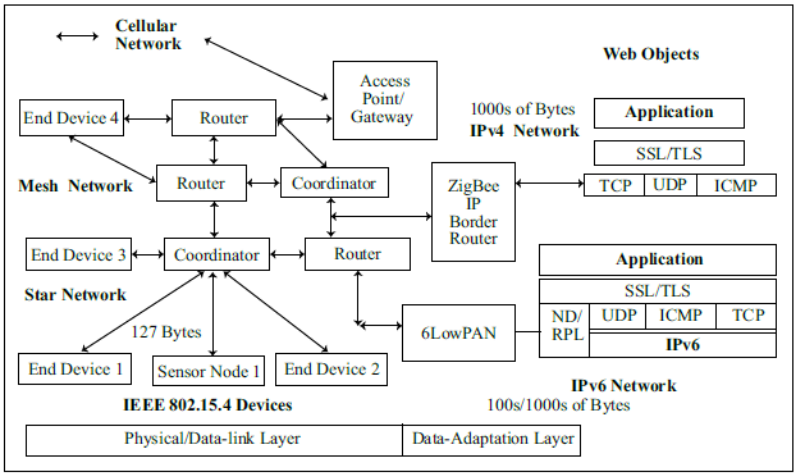
3.2 Wired Communication Technology:

- **UART/USART Serial Communication**: Ideal for serial communication
- **Serial Peripheral Interface (SPI)**: Four sets of signals, viz., SCLK, MISO, MOSI, and SS are used on four wires.
- **I2C Bus**: It has two lines that carry the signals, one is for clock, and another is for bidirectional data.

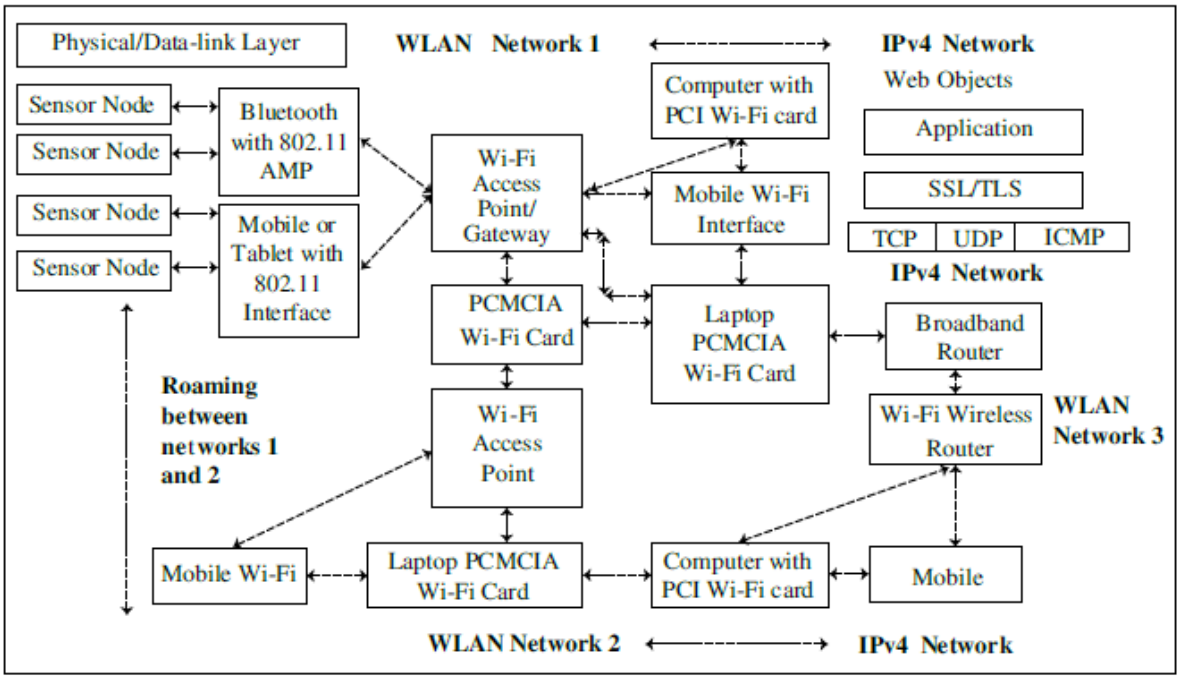
- **Wired USB**: It is used for fast serial transmission and reception between the hosts, the embedded system and distributed devices.
- **Ethernet**: This standard is widely known as IEEE 802.2 (ISO 8802.2) protocol for local area network of computers, workstations and device LANs.



(a)



(b)



(c)

Figure 1.6: (a) Connection between network and gateway using WPAN, (b) ZigBee end point, router, forming the mesh with IPv4 and IPv6 networks, and (c) three WLAN networks for sensor device nodes, mobile and computers and internet connectivity with the IP4 networks.

Property	NFC	BT LE	ZigBee IP	WLAN 802.11
IEEE Protocol		802.15.1	802.15.4	802.11z
Physical Layer	848, 424, 212, 106 kbps	2.4 GHz (LE-DSSS)	2.4 GHz or 915 MHz, 868 MHz and 433 MHz DSSS MAC layer CSMA/ CA	2.4 GHz Two PHY layers MAC layer CSMA/CD
Data Transfer Rate	106 kbps	1 Mbps	250 kbps (2.4 GHz, 40 kbps 915 MHz, 20 kbps 868.3 MHz	11 Mbps/54 Mbps
Form Factor and Range	10–20 cm	Small	Small 10 m to 200 m	Bigger
Protocol Stack		Small in LE	127 B	Bigger than WPAN devices
Power Dissipation	Very low	Lower than ZigBee, much lower than WLAN 802.11	2 mW Router and 0.1 mW for end-device Much lower than WLAN 802.11	Much Higher than ZigBee
Set up/ Connection/ Disconnection Intervals	0.1 s	3s Connection time < 3 ms	20 ms Connection time < 10 ms	
Security	—	AES-CCM-128	AES-CCM-128	WEP
Applications	Payment wallet, short distance communi-cation	WPAN, IoT/M2M devices, widely present in mobiles and tablets and, need addition circuit in sensors, actuators, controllers and IoT devices	WPAN, wider presence in sensors, actuators, controllers, automobile and medical electronic and IoT devices connectivity using IPv6, 6loWPAN, ROLL, RPL and TLSv1.2	WLAN and WWAN network tablet, desktops, mobiles, devices with PCMCIA interface, home networking, Easy IPv4 connectivity
Network	Point to point between active and passive devices	Star topology, peer-to-peer piconet expended by inter-piconets data transactions and synchronisation	Low power, mesh or peer-to peer star networks using end devices, coordinator, router, ZigBee IP border router	LAN topology IBSS, BSS and distributed BSSs for WWLAN widely used for Internet connectivity of mobiles, tablets, desktops
Network Characteristics	P2P mode, card emulation mode and reader mode Passive neighbour activation	Self-configuring, self-healing, self-discovery	Self-configuring, self-healing, self-discovery	Scalable, interoperability, security, integrity and reliability
Broadcast/ Multicast/Unicast	Unicast	Unicast	Unicast/ multicast	Unicast

Figure 1.7: Comparison of communication technologies.

4. Data Enrichment, Consolidation, and Device Management at Gateway

4.1 Data Enrichment:

This process adds value to raw data collected from sensors by appending contextual information such as location, timestamp, or device ID. Enriched data helps in better analysis and decision-making.

4.2 Data Consolidation:

Involves aggregating data from multiple sources, filtering redundant or noisy information, and packaging it for efficient transmission. Gateways often perform initial processing to reduce the burden on cloud infrastructure.

4.3 Device Management at Gateway:

Gateways serve as intermediaries between end devices and the cloud. They perform the following functions: - Device authentication and registration - Remote firmware updates - Health monitoring of devices - Managing connectivity and communication protocols

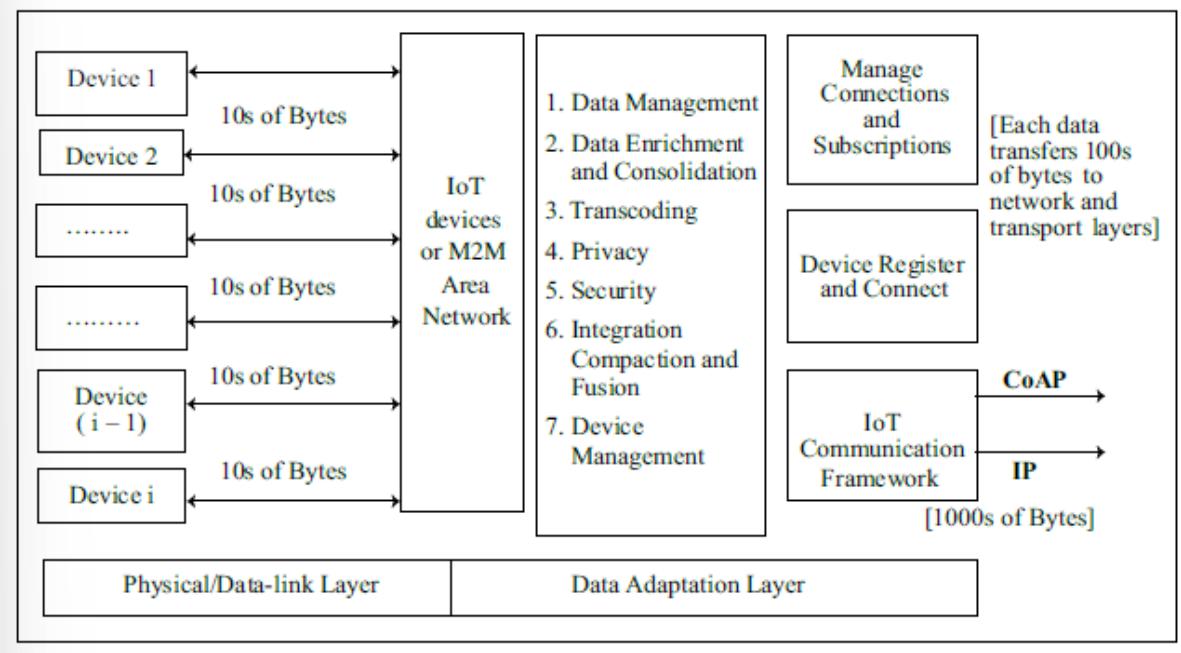


Figure 1.8: IoT gateway consisting of data enrichment and consolidation, device management and communication frameworks.

5. IoT and M2M (Machine-to-Machine)

5.1 M2M Communication:

M2M refers to the direct communication between two devices without human intervention. It is often used in industrial automation, telemetry, and logistics systems.

5.2 Differences Between IoT and M2M:

Feature	M2M	IoT
Communication Scope	Point-to-point	Network of devices over the internet
Network Type	Cellular, Wired	Wired, Wireless (Wi-Fi, Bluetooth, 5G, etc.)
Data Handling	Localized	Cloud-based with advanced analytics
Application Areas	Industrial automation	Smart homes, cities, health, agriculture, etc.
Scalability	Limited	Highly scalable

5.3 SDN and NFV for IoT:

- **SDN (Software Defined Networking)** separates the control plane from the data plane, enabling programmable network configuration. In IoT, SDN enhances network flexibility and supports dynamic routing for data.
- **NFV (Network Function Virtualization)** decouples network services from hardware, allowing them to run on virtual machines. It enables cost-efficient deployment of network functions like firewalls, load balancers, etc.
- These technologies support the growing demand for bandwidth, device density, and reliability in IoT systems.

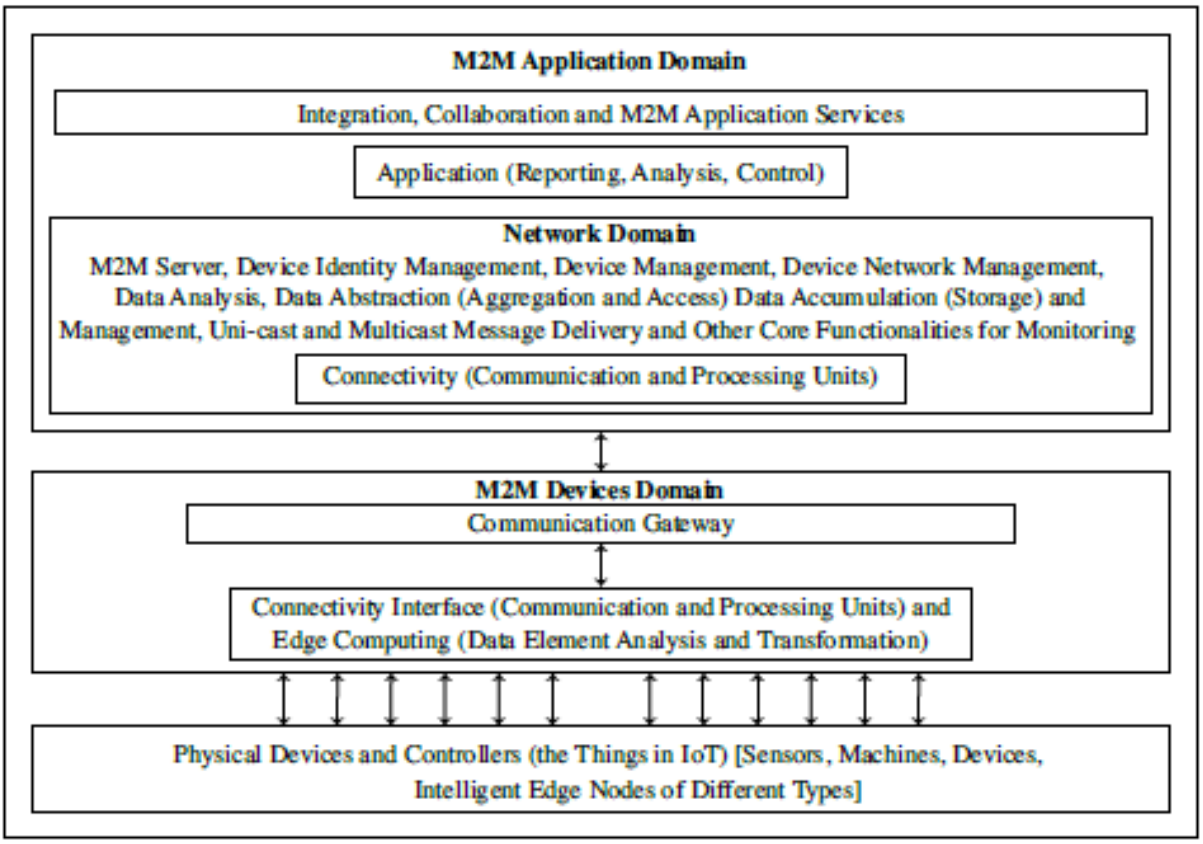


Figure 1.9: Domains of M2M architecture.

5.4 Software development tools

Examples of M2M software and development tools are as follows:

- **Mango** is an open source M2M web-based software. It supports multiple platforms, multiple protocols, databases, meta points, user-defined events and import/export.[20]
- **Mainspring** from M2MLabs is a development tool, and source framework for developing M2M applications.[21] It enables:
  - Flexible modelling of devices and their configurations
  - Communication between devices and applications
  - Validation and normalisation of data
  - Long-term data storage and data retrieval functions
  - Programming in Java and Apache Cassandra
  - Usages of no SQL database.
- **DeviceHive** is an M2M communication framework. It is an M2M platform and integration tool. It enables connecting devices to the IoT. It includes web-based management software that creates security-rules-based e-networks and monitoring devices. The web software enables prototype projects built with DeviceHub and online tests to find out how it works.
- **Open M2M Protocols, Tools and Frameworks:** Following are the open protocols, tools and frameworks for M2M:
  - XMPP, MQTT-OASIS standards group and OMA LWM2M-OMA standard group for protocol
  - Various projects of Eclipse M2M industry working groups' are Koneki, Eclipse SCADA for open standards for communication protocols, tools and frameworks
  - ITU-T Focus Group M2M global standardisation initiative for a common M2M service layer
  - 3GPPP study group for security aspects of M2M equipment and automatic SIM activation covering remote provisioning and change of subscription.
  - Weightless (wireless communications) group for standards and using wireless spaces for M2M

6. Domain Specific IoTs

6.1. Home Automation:

IoT transforms traditional homes into smart homes by enabling remote monitoring and control of household devices. Applications include: - Smart lighting and HVAC systems - Automated door locks and surveillance cameras - Voice-controlled assistants like Amazon Alexa and Google Assistant

6.2. Smart Cities:

Smart cities integrate IoT technologies for efficient management of resources and services. Examples include: - Smart traffic lights and real-time traffic monitoring - Waste collection systems with fill-level sensors - Smart Street lighting with motion sensors and energy optimization - Environmental monitoring for pollution and noise levels

6.3. Environment:

IoT devices are used to monitor and protect the environment through real-time data collection. Applications include: - Forest fire detection using temperature and smoke sensors - Air and water quality monitoring stations - Weather forecasting and disaster alert systems - Wildlife tracking and habitat protection using GPS collars

6.4. Health and Lifestyle:

Healthcare and fitness have significantly benefited from IoT technologies. Examples include: - Wearables like smartwatches and fitness bands for heart rate and activity monitoring - Remote patient monitoring systems for chronic disease management - Smart beds and devices for elderly care and fall detection - Mobile health applications and telemedicine platforms